An Explainable Machine Learning Framework for Early Multi-Disease Risk Prediction Using Healthcare Data Analytics

Mandha Varshitha
Department of Computer Science and Engineering
KLH University
Hyderabad, India

Chenna Keerthana
Department of Computer Science and Engineering
KLH University
Hyderabad, India

Edigi Supraja
Department of Computer Science and Engineering
KLH University
Hyderabad, India

Chintalapati Bhargavi Devi
Department of Computer Science and Engineering
KLH University
Hyderabad, India

***Abstract*—Early identification of disease risk can support preventive healthcare, but machine-learning systems are often developed as isolated disease-specific predictors and may provide limited insight into why a prediction is produced. This paper presents an explainable multi-disease healthcare analytics framework implemented as an end-to-end software prototype. The framework evaluates four conditions—diabetes, heart disease, chronic kidney disease (CKD), and thyroid cancer recurrence—using disease-specific structured datasets. Five candidate classifiers, namely logistic regression, decision tree, random forest, XGBoost, and Gaussian Naive Bayes, are trained under a common 80/20 stratified evaluation protocol, and the model with the highest receiver operating characteristic area under the curve (ROC-AUC) is selected for deployment for each condition. The framework additionally supports medical-report text extraction and optical character recognition (OCR), clinical-parameter parsing, patient-level SHAP and LIME explanations, a composite 0–100 health-risk index, and research-oriented decision support through a Flask web application. On the held-out test sets, the selected models achieved ROC-AUC values of 0.830 for diabetes (XGBoost), 0.960 for heart disease (random forest), 1.000 for CKD (random forest; tied with XGBoost), and 0.995 for thyroid cancer recurrence (random forest). The results demonstrate the feasibility of integrating model comparison, explainability, report processing, and multi-condition risk presentation in one workflow. The findings are preliminary and should not be interpreted as clinical validation; external validation on independent healthcare cohorts is required before clinical deployment.**

***Index Terms*—healthcare analytics, machine learning, multi-disease prediction, explainable AI, SHAP, LIME, chronic kidney disease, clinical decision support, Flask, OCR**

I. Introduction

Healthcare organizations increasingly generate heterogeneous clinical information in both structured datasets and unstructured medical reports. Machine learning (ML) can identify statistical patterns in such data and estimate disease risk before overt clinical deterioration. However, a practical healthcare prediction system requires more than a high-performing classifier: it must also support model comparison, interpretable outputs, transparent handling of missing information, and an interface through which predictions can be reviewed responsibly.

Recent studies have investigated multi-disease prediction, explainable artificial intelligence (XAI), and disease-specific clinical risk models. Men *et al.* developed a multi-disease prediction approach based on long short-term memory networks and highlighted its potential for intelligent clinical decision support [1]. Rao *et al.* compared LIME and SHAP as explainers for autonomous disease prediction systems and emphasized interpretability as a requirement for healthcare decision support [2]. Ghosh and Khandoker compared five ML methods for CKD prediction and used both SHAP and LIME to interpret the selected model [3]. Similarly, Ye *et al.* developed an interpretable model for several diabetes complications using multiple classifiers and SHAP [4]. More recent work continues to combine ML and XAI for disease prediction and clinical transparency [6]–[8].

Despite these advances, a gap remains between a collection of individual disease classifiers and a unified application workflow that handles heterogeneous inputs, compares multiple candidate models, selects disease-specific models, generates patient-level explanations, synthesizes multi-condition risk, and presents the outputs through an interactive decision-support interface. The implemented system studied in this paper addresses this integration problem as a research prototype.

The main contributions are:

- an integrated framework covering four disease/condition prediction modules: diabetes, heart disease, CKD, and thyroid cancer recurrence;
- a common model-selection procedure comparing logistic regression (LR), decision tree (DT), random forest (RF), XGBoost, and Gaussian Naive Bayes (NB) using ROC-AUC;

- an input-processing pipeline that can extract information from PDF/image medical reports and transform identified clinical parameters into model-ready inputs;
- patient-level explainability using SHAP and LIME, with language explicitly framed as model contribution rather than causal medical reasoning;
- a composite 0–100 research-prototype Health Risk Index and rule-based, non-prescriptive decision-support summary; and
- a Flask-based web application that connects prediction, explanation, risk synthesis, visualization, and persistence through SQLite.

II. Literature Survey

A. ML-Based Disease Prediction

Traditional statistical models remain useful for healthcare classification because they are relatively simple to train and interpret. At the same time, tree ensembles and gradient boosting can model nonlinear feature interactions. Ghosh and Khandoker evaluated LR, RF, DT, NB, and XGBoost for CKD and selected the best model using predictive performance measures including AUC [3]. Their use of SHAP and LIME demonstrates how model performance and interpretability can be evaluated together.

Ye *et al.* compared LR, DT, RF, and CatBoost for multiple diabetes complications and used SHAP to expose influential factors [4]. This supports the use of model comparison rather than assuming a single algorithm is optimal for every clinical problem.

B. Multi-Disease Prediction

Men *et al.* investigated multi-disease prediction using an LSTM architecture and emphasized the value of unified prediction for intelligent clinical decision support [1]. The present work differs in emphasis: rather than a temporal deep-learning architecture, it uses disease-specific structured classifiers and integrates them into a common application workflow.

A recent IEEE study also investigated a unified multi-disease prediction system with disease-specific classifiers and a web interface [9]. This reinforces the relevance of integrated multi-disease systems while also indicating that novelty should be claimed from the exact workflow and implementation rather than from the generic idea of multi-disease prediction alone.

C. Explainable AI in Healthcare

Rao *et al.* examined LIME and SHAP for autonomous disease prediction and argued that explanations can improve transparency of model-based decision support [2]. The broader interpretability literature has also evaluated SHAP, LIME, and other post-hoc methods across several medical classifiers and datasets [5]. Agrawal *et al.* combined DT, NB, RF, and XGBoost with SHAP and LIME for several disease prediction tasks [6]. Chouit *et al.* further demonstrated an interpretable CKD framework using XGBoost, SHAP, and LIME on hospital and UCI datasets [8].

D. Literature Comparison and Research Gap

Table I summarizes representative recent studies. The comparison distinguishes prior literature from the experimental baseline models used in the present project.

The literature indicates three recurring limitations relevant to the present project: (i) many studies focus on one disease or one disease family; (ii) prediction and explanation are frequently presented as separate experimental components; and (iii) deployment-oriented workflows may not combine report extraction, multi-condition prediction, explainability, and a unified risk summary. The present work therefore targets integration rather than claiming novelty for any individual ML algorithm.

III. Research Gap and Problem Definition

Existing healthcare ML research demonstrates strong predictive capability, but practical adoption can be constrained when models are isolated, opaque, or disconnected from the data-entry workflow. In particular, a user may have information in a medical report rather than in a structured model form. A unified system should therefore (1) extract usable parameters where possible, (2) determine whether sufficient disease-specific information exists, (3) run appropriate disease models, (4) expose important model contributions, and (5) summarize the results without presenting them as a clinical diagnosis.

The research problem addressed in this study is defined as follows: given heterogeneous patient information X, construct a set of disease-specific prediction functions $f_d(X_d)$ for four conditions d, select the strongest candidate model for each condition under a fixed validation protocol, explain the selected predictions, and synthesize the available probabilities into a transparent research-prototype risk index.

IV. Proposed Methodology

A. Overall Framework

Fig. 1 shows the implemented architecture. The workflow begins with structured datasets or a patient medical report. For report-based input, PDF/image text is extracted and clinical parameters are parsed and validated. Disease-specific feature vectors are then passed through the corresponding stored preprocessing pipelines and selected classifiers. The resulting probabilities are explained with SHAP and LIME when evaluation is possible. Finally, the application generates a composite research risk index and non-prescriptive decision-support observations.

B. Datasets

Four local structured datasets are included in the implementation. The diabetes dataset contains 768 rows and eight predictive features. The heart dataset contains 303 rows and 13 features. The CKD dataset contains 400 rows and 13 features. The thyroid dataset contains 383 rows and 16 features, with recurrence as the target. Table II summarizes the stored metadata and held-out splits.

The class counts recorded by the implementation are 500/268 for diabetes, 164/139 for heart disease, 150/250 for

TABLE I
RECENT LITERATURE COMPARISON (2021–2026)

Year	Study	Focus	Models / Methods	XAI	Relevance / Limitation Addressed
2021	Men *et al.* [1]	Multi-disease prediction	LSTM	Not the main focus	Unified multi-disease clinical decision support; temporal deep learning rather than the structured disease-specific workflow used here.
2022	Rao *et al.* [2]	Autonomous disease prediction	ML disease prediction analysis	SHAP, LIME	Directly motivates transparent prediction explanations.
2024	Ghosh and Khandoker [3]	CKD prediction	LR, DT, RF, NB, XGBoost	SHAP, LIME	Strong disease-specific XAI study; focuses on CKD rather than a four-condition application.
2024	Ye *et al.* [4]	Diabetes complications	LR, DT, RF, CatBoost	SHAP	Demonstrates interpretable complication risk modelling; scope is diabetes complications.
2024	Martinez *et al.* [5]	Medical model interpretability	Multiple classifiers	SHAP, LIME and other techniques	Establishes the need to evaluate global/local interpretability in medical ML.
2025	Agrawal *et al.* [6]	Multi-condition disease prediction	DT, NB, RF, XGBoost	SHAP, LIME	Closely related integrated XAI workflow; uses different disease set and implementation.
2025	Wu *et al.* [7]	CKD progression in T2DM	10 ML algorithms, XGBoost selected	SHAP	Demonstrates clinically oriented XAI and web deployment, with external validation.
2026	Chouit *et al.* [8]	CKD prediction	XGBoost compared with LR, RF, DT, NB	SHAP, LIME	Very recent CKD XAI benchmark; motivates careful validation and overfitting control.

TABLE II
DATASET SUMMARY

Condition	Rows	Features	Train	Test
Diabetes	768	8	614	154
Heart disease	303	13	242	61
CKD	400	13	320	80
Thyroid recurrence	383	16	306	77

CKD, and 275/108 for thyroid recurrence. No external clinical cohort is used in the current experiment.

C. Data Preprocessing

The training pipeline uses an 80/20 stratified train-test split with random state 42. For the diabetes pipeline, physiologically invalid zero values in designated fields are first converted to missing values, followed by median imputation and standardization. The general preprocessing pipeline uses median imputation followed by standard scaling. The thyroid module additionally applies a categorical preprocessing pipeline before model fitting. All preprocessing objects are fitted on the training partition and stored for inference.

D. Candidate Models and Selection

Five classifiers are compared for each condition:

1) Logistic Regression (LR), with maximum iterations of 1000;
2) Decision Tree (DT), using constrained depth and leaf-size parameters;
3) Random Forest (RF), using 150 trees and constrained depth/leaf-size settings;
4) XGBoost, using 100 estimators, depth 4, learning rate 0.08, subsampling, and column subsampling; and
5) Gaussian Naive Bayes (NB).

For every candidate, predictions and probabilities are generated on the held-out test partition. The deployment candidate is selected using ROC-AUC:

$$M_d^* = \arg\max_{m \in \mathcal{M}} AUC_{d,m}, \tag{1}$$

where $\mathcal{M}$ is the five-model candidate set and d denotes the disease condition. The selected model and preprocessing object are serialized for application inference.

E. Disease-Specific Prediction

For a selected classifier M_d^*, the predicted probability of the positive class is

$$p_d = P(y_d = 1 \mid X_d; M_d^*). \tag{2}$$

The prototype uses a threshold of 0.5 to produce a binary risk flag for the application interface. A probability is presented as a model output and not as a clinical probability of disease in an individual patient.

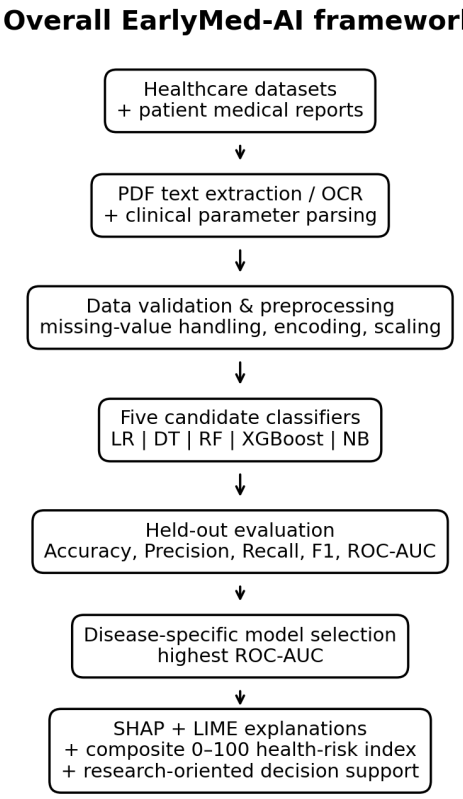


Fig. 1. System architecture of the explainable multi-disease healthcare analytics framework. The workflow includes input sources, report extraction and validation, disease-specific preprocessing, candidate-model evaluation and selection, prediction and explanation, risk synthesis, and application delivery.

The implemented application also derives certain proxy inputs when possible. For example, fasting glucose is derived from HbA1c using the implemented estimated-average-glucose relation when direct glucose is absent. For heart disease, fasting blood sugar is derived from a glucose threshold. Such derived values are documented here because they are part of the deployed inference workflow and should be considered when interpreting results.

F. Medical Report Processing

The application accepts PDF or image-based reports. Extracted text is passed through a clinical parameter parser and validation stage. Identified values are normalized into internal parameter representations. The system then checks disease-specific readiness before prediction. If required information is insufficient, the disease module returns an "Not Evaluated" status instead of forcing a prediction. This is a key safety-oriented design decision in the prototype.

G. Explainability with SHAP and LIME

The system uses SHAP for patient-specific feature contribution analysis. For tree-based selected models, a TreeExplainer is used when available. The contribution for a feature is interpreted operationally as movement toward a higher or lower model-estimated risk output. The application sorts contributions by absolute magnitude and exposes the top contributors.

LIME is used as a local surrogate explanation. A perturbation-based local explainer is created around the processed patient vector, and the highest-weight local rules are returned. SHAP and LIME are therefore complementary: SHAP provides additive feature contributions for the selected model, while LIME provides a local surrogate representation of the prediction behavior.

H. Composite Health Risk Index

When more than one disease probability is available, the prototype synthesizes the available probabilities into a 0–100 research index. Let $P = \{p_1, p_2, \ldots, p_k\}$ be the set of valid disease probabilities. The implemented calculation is

$$\bar{p} = \frac{1}{k} \sum_{i=1}^{k} p_i, \tag{3}$$

$$p_{max} = \max_i p_i, \tag{4}$$

$$R = 100(0.6 p_{max} + 0.4 \bar{p}). \tag{5}$$

The final score is rounded and bounded to [0,100]. The interface labels scores below 40 as lower estimated risk, 40–69 as moderate estimated risk, and 70 or above as higher estimated risk. This is explicitly an academic research-prototype score and not a clinical severity index.

I. Decision Support and Web Application

The Flask application connects report processing, validation, prediction, explanation, visualization, and persistence. SQLite is used to store report, extraction, prediction, and explanation records. The decision-support component generates observations based on model outputs and top explanation contributors. Its messages are intentionally non-prescriptive: it does not diagnose patients, prescribe medicines, or alter treatment plans. It directs users to correlate model outputs with clinical history, examination, and professional medical assessment.

J. Exploratory Data Analysis (EDA)

Exploratory data analysis was used to inspect the tabular datasets before interpreting the predictive results. The purpose of EDA in this project is not to establish clinical causation, but to understand dataset composition, target distribution, feature types, missingness, invalid values, and potential sources of evaluation bias. These checks are particularly important because the four datasets originate from different prediction tasks and use different feature definitions.

1) Dataset size and feature structure: The diabetes dataset contains 768 observations and eight predictors, the heart-disease dataset contains 303 observations and 13 predictors, the CKD dataset contains 400 observations and 13 predictors, and the thyroid recurrence dataset contains 383 observations and 16 predictors. The sample sizes are modest relative to the complexity and diversity of real healthcare populations. Consequently, one or two observations can materially affect the estimated performance on a small test set. The datasets are evaluated independently; their records are not pooled because their targets and feature semantics differ.

2) Target-class distribution: The target counts recorded in the project metadata are 500/268 for diabetes, 164/139 for heart disease, 150/250 for CKD, and 275/108 for thyroid recurrence. The class distributions are not identical. The CKD dataset contains more positive than negative records under the project target coding, whereas the thyroid dataset contains more non-recurrence than recurrence records. This variation makes accuracy alone insufficient for comparing classifiers. Precision, recall, F1-score, ROC-AUC, and—where available—precision-recall AUC provide complementary views of the classification behaviour.

3) Diabetes feature inspection: The diabetes features include pregnancies, glucose, diastolic blood pressure, skin thickness, insulin, body-mass index, diabetes pedigree function, and age. Several fields have domain-specific constraints. In the implementation, zero values in glucose, blood pressure, skin thickness, insulin, and BMI are treated as invalid or missing rather than as physiologically meaningful measurements. This step is important because treating an invalid zero as a genuine measurement can distort the distribution and the decision boundary. The project then applies imputation and scaling in the model pipeline. Derived estimated average glucose (eAG), when displayed by the application, is computed from HbA1c using the approximation $eAG = 28.7 \times HbA1c - 46.7$; it is a derived presentation value and is not an additional trained feature unless explicitly present in the model input.

4) Heart-disease feature inspection: The heart dataset combines demographic variables, symptom/category codes, resting blood pressure, cholesterol, fasting blood sugar, resting electrocardiographic category, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, and thalassemia-related category. These variables have mixed scales and mixed measurement types. Numeric scaling and consistent category handling therefore matter. Category codes should not automatically be interpreted as continuous clinical quantities; the meaning of a coded feature depends on the original dataset documentation.

5) CKD feature inspection: The CKD dataset contains age, blood pressure, blood glucose, blood urea, serum creatinine, haemoglobin, packed cell volume, white-cell count, red-cell count, albumin, specific gravity, hypertension, and diabetes-related fields. These measurements combine laboratory values with categorical or binary clinical indicators. Missingness is an important concern for this dataset family, and preprocessing must be fitted only on the training partition to prevent information from the test partition entering the learned transformation. The exceptionally high stored CKD scores warrant a separate audit of duplicated records, target-correlated fields, preprocessing leakage, and split overlap before any broad generalization claim is made.

6) Thyroid recurrence feature inspection: The thyroid recurrence dataset includes age, gender, smoking history, prior radiotherapy, thyroid function, physical examination, adenopathy, pathology, focality, risk category, tumour classification fields, stage, response, and recurrence label. Categorical fields are transformed using one-hot encoding within the preprocessing pipeline. The resulting transformed feature space is larger than the original 16-column feature table. This is expected when categories are represented as indicator columns, but it also means the effective model input dimension should be reported separately from the raw dataset column count when discussing computational cost.

7) Missing values, invalid values, and encoding: The preprocessing design combines imputation for missing numeric values, scaling where required, and one-hot encoding for thyroid categorical variables. For valid evaluation, preprocessing transformations must be learned from the training subset and then applied unchanged to the test subset. Input validation in the application checks whether extracted or manually entered values are usable before inference. A PDF/OCR pipeline can introduce a second type of missingness: a clinical measurement may be present in the report but not correctly recognized by the text extraction system. These extraction errors should be audited separately from missing cells in the benchmark CSV files.

8) Distribution checks and descriptive summaries: A complete EDA workflow should begin with a column-level inventory containing data type, valid range, missing count, unique-value count, and summary statistics. Numeric columns should be summarized using the mean, standard deviation,

median, interquartile range, minimum, and maximum. Categorical columns should be summarized with frequency tables and the proportion of each category. This dual summary is necessary because a numeric data type does not guarantee that a variable is continuous: some heart-disease and thyroid fields are category codes whose values represent labels rather than meaningful magnitudes. In plots, numeric variables are best inspected with histograms and box plots, while categorical variables are best inspected with ordered bar charts. Each plot should include the sample count and a readable axis label so that apparent differences are not interpreted without knowing the underlying number of observations.

The present source package records dataset dimensions and target counts, but it does not provide a complete set of verified feature-level descriptive statistics for all four datasets. For that reason, this paper avoids reporting unverified means, medians, correlation coefficients, or outlier counts. In the final reproducible experiment, these summaries should be generated from the exact CSV files used by the training scripts and stored alongside the dataset checksum. The EDA output should also record the target coding convention, because a positive label in one benchmark need not represent the same clinical event as a positive label in another.

9) Class balance and metric implications: The recorded class counts show that the datasets differ in their target balance. A majority-class classifier could therefore achieve different baseline accuracy values across tasks even if it had no ability to discriminate individual risk. For this reason, accuracy is interpreted alongside precision, recall, F1-score, and ROC-AUC. Recall is particularly relevant when false negatives are important, while precision describes the fraction of positive predictions that are correct under the test-set label definition. ROC-AUC evaluates ranking across thresholds; it does not identify a clinically appropriate decision threshold by itself. Since the paper's model-selection rule uses ROC-AUC, threshold-specific results should be interpreted separately and not as a direct consequence of the selection metric.

10) Missingness and potential data-quality mechanisms: Missing values may arise for different reasons: a measurement may not have been collected, a value may be unavailable in the source record, or a data-conversion process may have encoded missingness as a special numeric value. These mechanisms are not interchangeable. In the diabetes dataset, zero values in glucose, blood pressure, skin thickness, insulin, and BMI are handled as invalid or missing by the implementation. This is a domain-informed preprocessing decision and should be checked against the original dataset documentation. For CKD and thyroid data, missingness and category normalization require special attention because inconsistent strings, whitespace, and mixed encodings can create artificial categories or silently increase the transformed feature space.

The report-ingestion component introduces additional quality checks that are distinct from dataset preprocessing. Optical character recognition can confuse characters such as "0" and "O", omit decimal points, or associate a value with the wrong label. Therefore, extracted values should retain their source text and be validated against expected units and plausible ranges before they are passed to a classifier. If extraction confidence is low or a required field is absent, the safer system behaviour is to request confirmation or report that the prediction cannot be completed, rather than silently substituting a potentially misleading value.

11) Relationships, redundancy, and leakage checks: Correlation matrices can help identify strongly associated numeric variables, but correlation should not be treated as evidence of causation and can be misleading for encoded categories. Pairwise plots and target-stratified distributions can be useful for detecting unusual separability, duplicate rows, and features that may directly or indirectly encode the target. The near-perfect CKD scores in the current stored results make these checks especially important. A leakage audit should examine duplicate records across the train/test split, columns derived from the outcome, preprocessing performed before splitting, and any target-related fields retained as predictors. The audit should be documented even if it finds no issue; without such evidence, near-perfect test results should remain provisional.

12) EDA outputs recommended for the final artifact: For reproducibility, the final project artifact should include one target-distribution chart per dataset, a missingness summary, representative numeric-feature histograms, and a table describing each feature's type and preprocessing rule. If the same EDA script is rerun after a dataset changes, the report should record the dataset version and row counts so that visual comparisons remain meaningful. The EDA figures should be generated from the actual project data rather than drawn schematically. Their role is to make the subsequent modelling decisions traceable: readers should be able to see which data characteristics motivated imputation, encoding, metric selection, and additional leakage checks.

13) EDA interpretation and limits: EDA helps identify data quality concerns and explains why the four tasks should be modelled separately. It does not demonstrate that a feature causes a disease or that the dataset represents the wider population. The current paper does not claim a complete clinical EDA audit with clinician-reviewed outlier labels, site-level distribution checks, or subgroup fairness measurements. These analyses remain useful extensions for a future study. For publication, dataset distribution plots, missing-value summaries, and feature histograms should be generated directly from the exact version of each dataset used in training and included with their counts and axis labels.

V. Experimental Setup and Evaluation Metrics

All reported model metrics are computed from the stored held-out test results generated by the project. The evaluation includes accuracy, precision, recall, specificity, F1-score, ROC-AUC, F_β, PR-AUC, log loss, and Matthews correlation coefficient (MCC). For a positive-class classifier, accuracy is

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{6}$$

Precision and recall are

$$Precision = \frac{TP}{TP + FP}, \qquad Recall = \frac{TP}{TP + FN}. \quad (7)$$

Specificity is

$$Specificity = \frac{TN}{TN + FP}, \quad (8)$$

and the F1-score is

$$F1 = 2\frac{Precision \cdot Recall}{Precision + Recall}. \quad (9)$$

ROC-AUC is the primary model-selection metric because it summarizes discrimination across probability thresholds. PR-AUC and MCC are also reported because class distributions differ between datasets.

VI. Results and Analysis

A. Model Performance

The four tables in this subsection are intended to be read as disease-specific comparisons rather than as one cross-disease leaderboard. Each dataset has a different target definition, feature space, sample size, and class balance; therefore, a metric value from one task cannot be directly interpreted as evidence that its model is intrinsically superior to a model trained for another task. Within each table, the rows compare the five candidate algorithms on the same held-out split. Bold values identify the best observed value for a metric in that task, while the deployment choice follows the paper's stated ROC-AUC selection rule. The discussion following the tables explains why these two views can differ.

Tables III–VI present the complete held-out test-set comparison. Percentages are shown for readability.

The diabetes table shows that the strongest value depends on the metric. Random Forest achieves the highest listed accuracy (76.62%), whereas XGBoost is selected because its ROC-AUC (83.00%) is the largest in the comparison. Thus, the selection rule favours ranking quality over accuracy at one threshold; it does not imply that XGBoost is best for every possible operational objective. Precision and recall should be examined together when considering how the model behaves at the saved classification threshold.

The heart-disease table shows stronger held-out discrimination for Random Forest, with 90.16% accuracy and 96.00% ROC-AUC. XGBoost and Naive Bayes have higher recall in the stored results, illustrating a trade-off: a model can identify a greater share of positive cases while also producing more false positives. Selection for a real workflow would require a prespecified operating objective and clinical input about the relative consequences of each error type; this prototype uses ROC-AUC as a consistent research comparison criterion.

For CKD, the results are unusually high and require cautious interpretation. XGBoost records perfect values for several listed metrics, while Random Forest and Naive Bayes also reach a ROC-AUC of 1.00. Accordingly, the table supports the statement that several candidates separate the held-out records extremely well, but it does not support a claim that Random Forest is uniquely best. Duplicate checks, target leakage checks, preprocessing review, and independent external validation are needed before these results can be generalized.

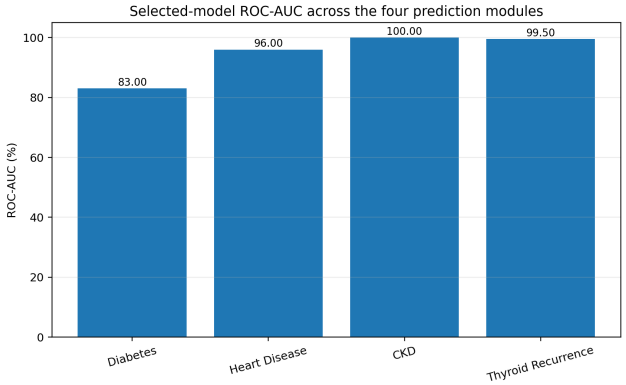


Fig. 2. ROC-AUC of the selected models across the four conditions.

For thyroid cancer recurrence, Random Forest has the highest ROC-AUC at 99.50%, while Logistic Regression and XGBoost show higher accuracy and recall in the stored comparison. The difference is small in ROC-AUC and should not be overstated, especially because the test set contains only 77 records. The results are best presented as evidence of strong ranking performance on this split, not as proof of clinical readiness or reliable recurrence detection in a new population.

B. Model Selection Findings

The ROC-AUC selection rule identifies XGBoost for diabetes, RF for heart disease, RF for CKD, and RF for thyroid recurrence. The diabetes result illustrates why a single model should not be assumed to be optimal: RF has the highest accuracy, but XGBoost has the highest ROC-AUC and is therefore selected by the implemented rule. For heart disease, RF achieves the highest ROC-AUC and accuracy among the tested candidates. CKD is an important edge case because RF and XGBoost both obtain ROC-AUC of 1.00; the stored implementation selects RF because it appears first among the tied maximum values. The thyroid module selects RF because its ROC-AUC (0.9950) is slightly higher than XGBoost (0.9942).

C. Confusion Matrix Analysis

The selected-model confusion matrices provide the counts of true-positive, true-negative, false-positive, and false-negative predictions for each held-out test set. They complement the threshold-independent ROC-AUC values by showing the outcomes at the implemented classification threshold. The four matrices are grouped in one figure to keep the presentation compact.

D. Comparative Performance Analysis

The performance plots compare the five candidate classifiers for each disease using accuracy, precision, recall, F1-score, and ROC-AUC. The four disease-specific plots are grouped into a single figure. The numerical values in Tables III–VI remain the primary source for exact metric comparisons.

TABLE III
DIABETES TEST-SET PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR	70.78	60.00	50.00	54.55	81.30
DT	73.38	68.57	44.44	53.93	81.19
RF	76.62	70.45	57.41	63.27	81.67
XGBoost	74.03	63.46	61.11	62.26	**83.00**
NB	70.13	56.67	62.96	59.65	76.46

TABLE IV
HEART DISEASE TEST-SET PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR	86.89	81.25	92.86	86.67	95.13
DT	78.69	72.73	85.71	78.69	88.47
RF	**90.16**	**86.67**	92.86	**89.66**	**96.00**
XGBoost	88.52	81.82	**96.43**	88.52	94.81
NB	86.89	79.41	**96.43**	87.10	95.24

TABLE V
CHRONIC KIDNEY DISEASE TEST-SET PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR	97.50	98.00	98.00	98.00	99.93
DT	97.50	98.00	98.00	98.00	99.80
RF	98.75	98.04	100.00	99.01	100.00
XGBoost	**100.00**	**100.00**	**100.00**	**100.00**	100.00
NB	95.00	100.00	92.00	95.83	100.00

TABLE VI
THYROID CANCER RECURRENCE TEST-SET PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR	**97.40**	100.00	**90.91**	**95.24**	99.01
DT	94.81	95.00	86.36	90.48	96.94
RF	96.10	100.00	86.36	92.68	**99.50**
XGBoost	**97.40**	100.00	**90.91**	**95.24**	99.42
NB	93.51	94.74	81.82	87.80	97.89

E. Disease-Level Interpretation

For diabetes, XGBoost reaches 83.00% ROC-AUC and 74.03% accuracy, while RF obtains higher accuracy (76.62%) but lower ROC-AUC (81.67%). This difference highlights the value of reporting multiple metrics rather than relying on accuracy alone.

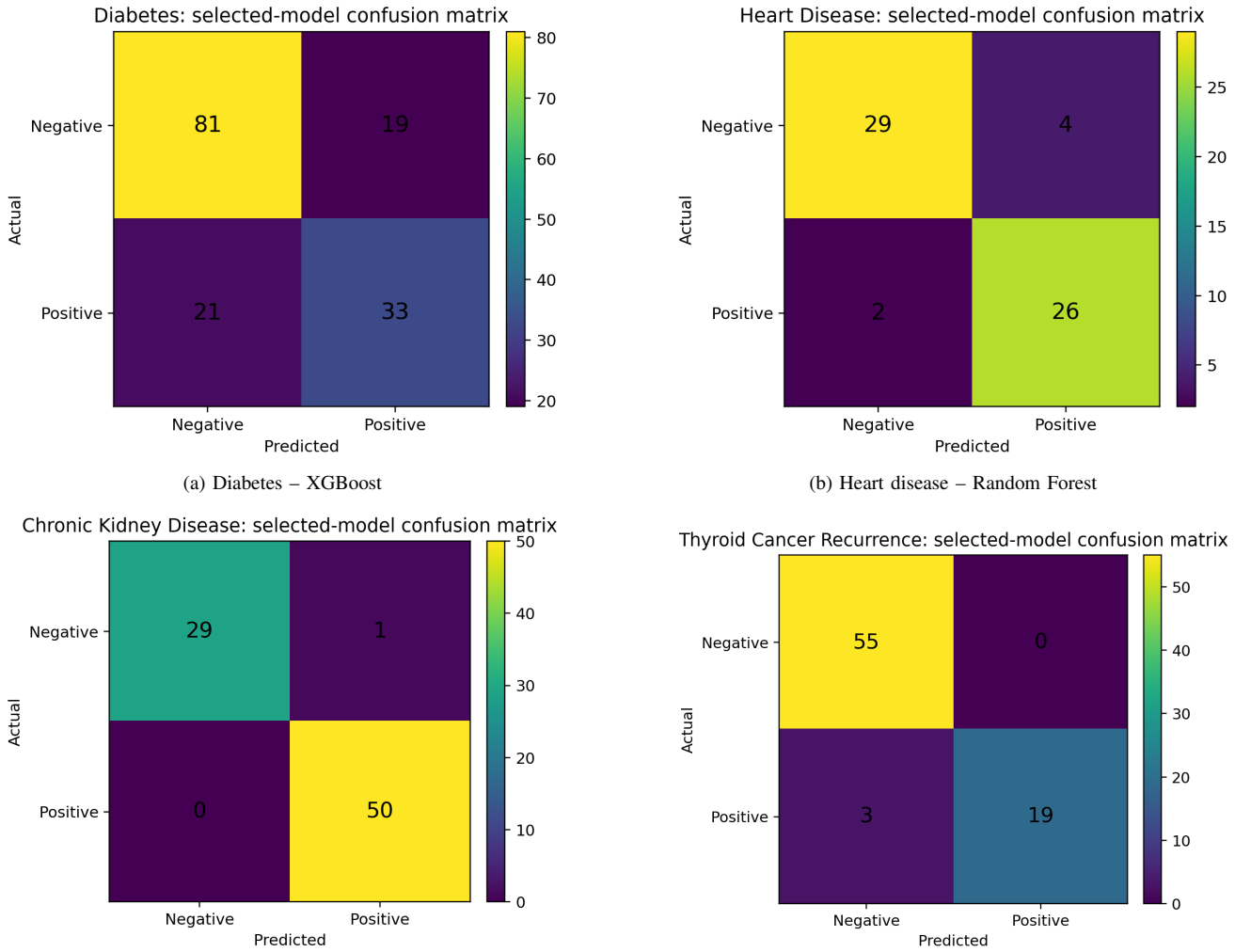
For heart disease, RF achieves 90.16% accuracy, 89.66% F1, and 96.00% ROC-AUC. XGBoost and NB obtain the highest recall at 96.43%, indicating that recall-sensitive use cases may lead to a different model preference even when ROC-AUC is the deployment criterion.

The CKD results are exceptionally high. XGBoost produces a perfect held-out confusion matrix in the stored experiment, while RF records one false positive and no false negatives. Because perfect performance can arise from dataset characteristics, feature separability, or leakage, these results must be treated as dataset-specific and should be verified with independent external data before any clinical interpretation.

For thyroid recurrence, LR and XGBoost have the highest accuracy (97.40%) and F1 (95.24%), while RF has the highest ROC-AUC (99.50%) and is selected. This again demonstrates the distinction between model selection and any single performance metric.

F. Explainability and Dashboard Output

The implemented system uses SHAP to show feature contributions and LIME to provide local surrogate explanations for individual predictions. These explanations describe model behaviour and should not be interpreted as causal or clinical evidence. The Flask dashboard brings together disease-specific predictions, the composite Health Risk Index, and explanation summaries. Figs. 5 and 6 show the patient-level explanations, while Fig. 7 shows the application interface.



(a) Diabetes – XGBoost

(b) Heart disease – Random Forest

(c) CKD – Random Forest

(d) Thyroid recurrence – Random Forest

Fig. 3. Confusion matrices of the selected disease-specific models on the held-out test sets.

VII. Discussion

The experimental results demonstrate that model ranking varies by disease and metric. This supports the use of disease-specific model selection rather than deploying one classifier universally. The selected models also show that strong discrimination can coexist with different recall and precision trade-offs. In a healthcare setting, such trade-offs should be evaluated against the intended screening or decision-support objective rather than optimized for accuracy alone.

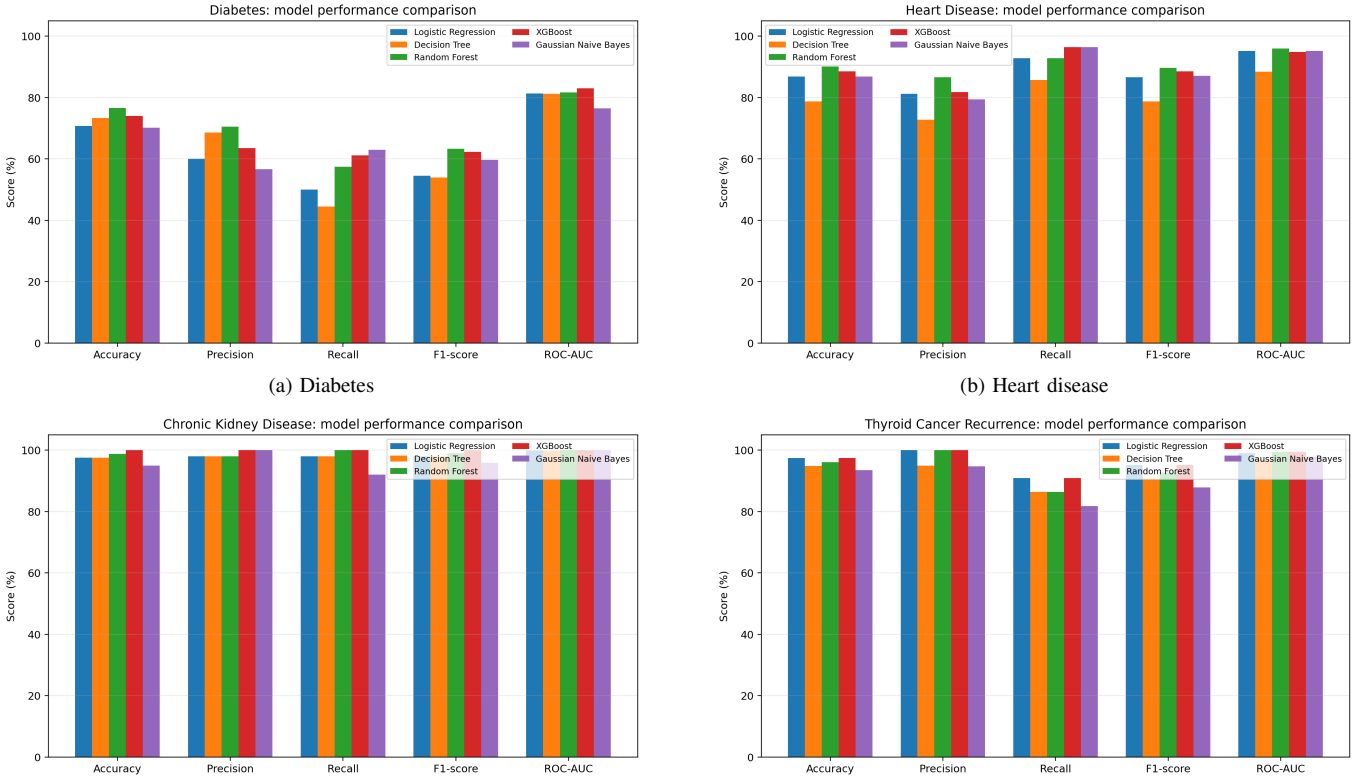
The framework's principal contribution is integration. The same application combines structured prediction, unstructured report extraction, disease-specific model selection, local explanations, risk synthesis, and a non-prescriptive decision-support layer. This integrated workflow is useful for demonstrating how separate ML components can be connected into a practical prototype.

However, the current evidence is not sufficient for clinical deployment. The datasets are relatively small, the evaluation is based on a single stratified hold-out split, and no independent external validation cohort is included. The CKD and thyroid results are particularly strong and should be investigated for dataset-specific separability, duplicate observations, feature leakage, and distributional artifacts. In addition, the report parser may impute or default missing thyroid categorical values in the deployed inference path; therefore, application-level predictions may not have the same information completeness as benchmark test-set predictions.

VIII. Novelty and Innovation

The novelty of the present work should be understood as a systems-level integration rather than invention of new classifiers. The framework contributes the following integrated workflow:

1) heterogeneous medical-report and structured-data input;
2) clinical parameter extraction and validation;
3) disease-specific comparison of five standard ML classifiers;
4) ROC-AUC-driven disease-specific model selection;
5) patient-level SHAP and LIME explanations;



(a) Diabetes

(b) Heart disease

(c) CKD

(d) Thyroid recurrence

Fig. 4. Disease-wise comparison of candidate classifiers across the reported evaluation metrics.



Fig. 5. Representative patient-level SHAP feature contributions generated by the implemented system.

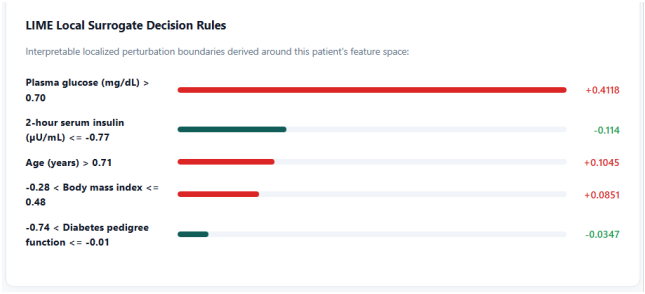


Fig. 6. Representative patient-level LIME local surrogate rules generated by the implemented system.

6) synthesis of available disease probabilities into a 0–100 research Health Risk Index; and
7) Flask-based presentation of predictions, explanations, and non-prescriptive decision-support observations.

This combination is the defensible contribution of the project. The individual algorithms, SHAP, LIME, OCR, and Flask are established technologies and are therefore not claimed as novel individually.

IX. Limitations

- The current evaluation uses a single 80/20 stratified hold-out split and does not report cross-validation confidence intervals.
- No independent external validation dataset is used.
- The datasets are relatively small and heterogeneous in feature semantics.
- The CKD and thyroid results require additional leakage and generalization checks because of near-perfect discrimination.
- Report extraction and OCR can introduce parsing errors; missing or ambiguous values can lead to imputa-

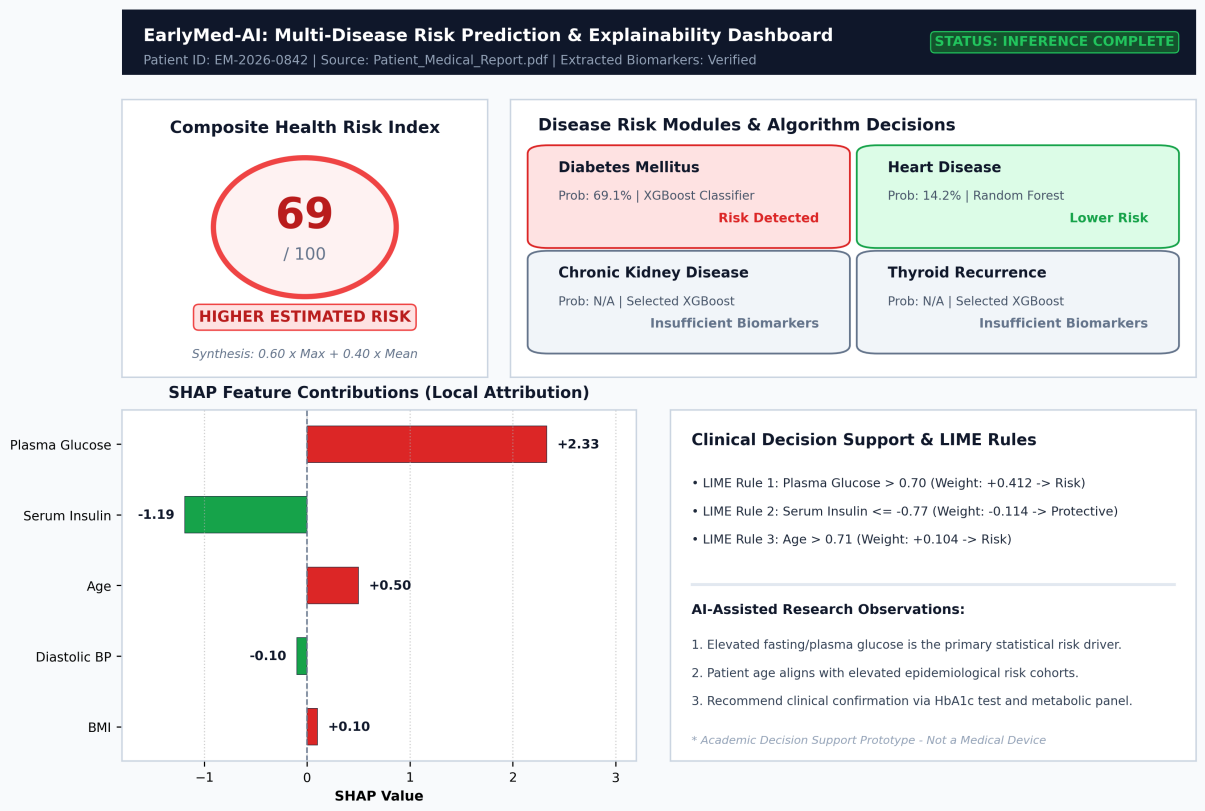


Fig. 7. Flask-based multi-disease prediction and decision-support dashboard.

tion/default handling.

- The composite Health Risk Index is a prototype mathematical synthesis and has not been clinically calibrated or validated.
- The decision-support text is research-oriented and must not be interpreted as diagnosis or treatment advice.

X. Future Scope

Future work should evaluate the framework on larger multi-center datasets and independent external cohorts. Repeated stratified cross-validation, calibration analysis, confidence intervals, sensitivity analysis, leakage checks, and subgroup fairness analysis should be added. The report-processing pipeline can be strengthened with domain-specific language models and human verification of extracted values. Future versions can also incorporate calibrated probabilities, temporal patient histories, clinician feedback, and prospective evaluation while preserving the distinction between decision support and medical diagnosis.

XI. SDG Alignment

The work supports Sustainable Development Goal 3 (Good Health and Well-Being) through early risk analytics and preventive decision support. It supports SDG 9 (Industry, Innovation and Infrastructure) through the development of an applied AI healthcare software framework. It can also contribute to SDG 10 (Reduced Inequalities) when future versions are validated across diverse populations and designed to reduce performance disparities across demographic groups.

Author Contributions

Edigi Supraja contributed to software testing, manuscript preparation, experimentation support, result verification, and quality review. Mandha Varshitha contributed to application development, user-interface integration, application workflow design, documentation, and integration of prediction and reporting components. Chenna Keerthana contributed to the literature review, results organization, explainability analysis, interpretation of SHAP and LIME outputs, and preparation of comparative analysis. Chintalapati Bhargavi Devi contributed to project implementation, data and model pipeline development, preprocessing integration, model evaluation workflow, and technical implementation documentation. All authors reviewed the manuscript and contributed to refining the final presentation of the research.

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